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Hybrid ensemble federated learning using SMOTE-Tomek for efficient DDoS detection on constrained edge devices over 5G networks

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ABSTRACT

The surge in connected devices within 5G edge computing ecosystems has intensified the threat of Distributed Denial-of-Service attacks, challenging the reliability and security of emerging networks. As 5G Phase2 introduces enhanced capabilities like standalone architecture, network slicing, ultra-reliable low-latency communication, and massive machine-type communication, traditional centralized machine learning models fall short due to privacy, latency, and scalability concerns. To address this, the proposed simulation introduces a hybrid ensemble detection framework grounded in federated learning principles, where three distinct models, XGBoost, Gradient Boosting, and Random Forest are deployed on simulated clients with localized, balanced data partitions. Data imbalance is mitigated using SMOTE (Synthetic Minority Oversampling Technique) and Tomek Links (Down sampling) technique, enhancing class distribution and model performance. Instead of sharing entire models, only the class probability outputs are transmitted for aggregation, significantly reducing communication overhead while preserving privacy. For ensemble, Soft Voting and Stacking with Logistic Regression as the meta-learner are used to classify attack traffic. Evaluations on the NCSRD-DS-5GDDos dataset, sourced from devices including Raspberry Pi 4 (Waveshare 5G Hat M2), Industrial 5G Router, Wi-Fi 6 Mobile Hotspot, and Customer Premises Equipment (CPE) Box, ensuring a realistic attack simulation over various network cells. It demonstrates high detection efficacy for DDoS types like UDP Flood, SYN Flood, HTTP GET Flood, ICMP Flood, and DNS Amplification. Achieving up to 98 % accuracy, precision, recall, F1-score, and 99.93 % AUC-ROC with minimal latency and computational load, the method ensures robust, privacy-preserving real-time detection in 5G/6G edge networks.

1. Introduction

The 5G network has emerged as a critical driver of Internet of Things (IoT) improvements, surpassing previous cellular restrictions to accommodate low-power, low-bandwidth devices. Compared to Long-Term Evolution (LTE), which was designed for mobile users, 5G has larger coverage, increased dependability, and ultra-low latency, making it perfect for massive machine-type communications (mMTC) and crucial IoT applications [1].The development of 5G technology has gone through several stages, as specified by 3GPP releases aimed at boosting connectivity, speed, and security. 5G Phase 1 (Release 15) established the New Radio (NR) standard, which enabled faster data speeds and greater spectrum efficiency. This phase emphasized enhanced mobile broadband (eMBB) and used a non-standalone (NSA) architecture, which relied on 4G LTE for control signaling and introduced network slicing [2].With 5G Phase2 (Release 16 & 17), the change to a stand-alone (SA) architecture eliminates LTE reliance, dramatically increasing Ultra-Reliable Low-Latency Communication and mMTC for IoT scalability. Enhanced network slicing permitted industry-specific customization, while edge computing enabled real-time data processing, lowering latency for applications such as self-driving vehicles [3].The future 5G Advanced (Release 18 and later) improves on these advances, adding AI-driven network optimization, increased energy efficiency, and advanced cyber security systems to tackle emerging threats. Furthermore, Non-Terrestrial Networks (NTN) are being developed to provide satellite-based 5G connection, providing global coverage [4]. In high-speed vehicular communication, 5G significantly outperforms 4G by offering ultra-low latency, enabling real-time data exchange. Its advanced architectural support and mobility management techniques ensure enhanced Quality of Service in dynamic, high-mobility scenarios

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[5].Modified Convolutional Neural Network (MCNN) enhances real-time threat detection and traffic management in 5G SDN by leveraging neural routing and deep pattern recognition[6].Despite its advancements, 5G-enabled IoT edge devices remain vulnerable to cyber security threats, requiring optimized attack detection models that balance performance and resource constraints [7]. These devices often operate with low processing capacity, restricted memory, and limited network resources, making them highly susceptible to Distributed Denial-of-Service (DDoS) attacks, unauthorized access, and anomaly-based intrusions. Modern DDoS attacks employ both high-volume flooding and low-volume stealth tactics, making them difficult to detect using traditional security methods [8]. Traditional Intrusion Detection Systems (IDSs) rely on centralized attack detection models, which struggle with latency, computational inefficiency, and data privacy risks, rendering them unsuitable for constrained 5G IoT edge devices. Federated Learning has surfaced as an innovative approach, facilitating decentralised model training on edge devices while eliminating the necessity for raw data transfer [9]. FL technique that incorporates DBSCAN clustering (FedDB) for enhanced DDoS attack detection also ensures privacy-preserving distributed learning, minimizing security vulnerabilities while enhancing attack detection performance [10]. Ensemble multiple models are combined in machine learning, a potent machine learning technique that enhances generalisation, accuracy, and resilience [11].

2. Motivation for this work

- 5G edge devices are highly vulnerable to DDoS attacks due to constrained resources.
- Centralized ML models pose privacy and latency challenges.
- Need for scalable, privacy-preserving detection frameworks.
- Class imbalance in attack data reduces detection accuracy.
- Existing FL models lack ensemble integration and real-time validation.

3. Contribution to the proposed work

The proposed research introduces a simulation-driven Ensemble Federated Learning (EFL) framework for detecting Distributed Denial of Service (DDoS) attacks in 5G edge environments. The framework integrates ensemble classifiers such as Random Forest, XGBoost, and Gradient Boosting [41] within a federated learning architecture that shares class probability vectors instead of model weights. This design reduces communication overhead and preserves data privacy, making it suitable for distributed, resource-constrained settings.

To improve detection accuracy and resilience, each participating node trains its ensemble models locally. Predictions are aggregated using Soft Voting and Stacking meta-learners, enabling robust classification of DDoS attacks versus benign traffic without relying on centralized coordination. This decentralized approach aligns with the operational realities of edge computing, where bandwidth and latency constraints limit the feasibility of global model synchronization.

The framework addresses the challenge of class imbalance in network traffic by incorporating Synthetic Minority Oversampling Technique (SMOTE) [12] and Tomek Links [13]. These techniques enhance the representation of minority attack types and reduce noise, improving the detection of rare but critical threats. Additionally, Recursive Feature Elimination (RFE) [14] is used to select the most relevant features, while Principal Component Analysis (PCA) [15] performs dimensionality reduction. Together, these steps improve computational efficiency and make the framework viable for deployment on edge devices with limited processing power.

All evaluations are conducted through realistic simulations that model edge device constraints and network conditions. The results demonstrate high detection accuracy, low latency, and robustness across heterogeneous traffic patterns. While the framework has not yet been implemented on physical devices, the simulation setup reflects practical constraints and validates its potential for future engineering applications in smart cities, industrial IoT, and remote healthcare systems.

Section 1 introduces the need for effective DDoS detection in 5G-enabled edge environments. Section 2 presents the motivation of the proposed work. Section 3 outlines the key contributions, including the development of a Hybrid Ensemble Federated Learning framework using Soft Voting and Stacking techniques. Section 4 provides the background, covering key concepts in federated learning, ensemble approaches, and cyber security applications in IoT and 5G systems. Section 5 reviews related works, identifying current trends and limitations in distributed attack detection. Section 6 details the proposed work, outlining data preprocessing, feature selection, and the design of the Hybrid Ensemble Federated Learning (EFL) framework using Soft Voting and Stacking methods. Section 7 explains the experimental setup, including dataset description, simulation environment, and hyper parameter configurations. Section 8 focuses on performance evaluation, presenting results through classification metrics, confusion matrices, system-level benchmarks, and ROC analyses to compare individual classifiers with Hybrid EFL variants. The primary results of the research are outlined, which also emphasises the accuracy and system effectiveness of the proposed soft voting and stacking-based Hybrid EFL model. Section 9 discusses practical engineering relevance and deployment considerations, addressing scalability, client heterogeneity, and real-time integration. Section 10 concludes the study by summarizing the findings and proposing future enhancements such as adaptive client weighting and multi-task learning.

4. Background

4.1. DDoS flooding attacks

Fig. 1 Illustrates the evolving trends in DDoS attacks based on Cloud flare's threat data, highlighting changes in attack distributions from 2021 to 2025. The prevalence of SYN floods has declined significantly, dropping from 54 % in 2021 to 30 % in 2025, indicating a shift away from this traditional attack strategy. In contrast, both HTTP attacks and UDP floods have surged in frequency, with HTTP attacks rising from 10 % to 30 % and UDP floods increasing from 20 % to 35 % over the same period. Meanwhile, ICMP and DNS amplification attacks have seen a slight reduction, reflecting changes in attacker methodologies as well as improvements in defensive measures. These trends underscore the dynamic nature of DDoS threats and the necessity for adaptable security mechanisms to combat emerging attack techniques [16].

A DDoS flooding attacks illustrated Fig. 2. That coordinated cyber attack in which a number of hacked devices produce excessive traffic to overwhelm a target, typically a server or website, triggering service degradation or total outage for authorized users. These attacks utilise the advantages of network weaknesses and can take a variety of forms. For example, UDP Flood sends a lot of User Datagram Protocol (UDP) packets; ICMP Flood uses Internet Control Message Protocol (ICMP) requests to overload network bandwidth; SYN Flood uses excessive TCP SYN requests to exhaust server resources; DNS Amplification Flood uses open DNS resolvers to magnify attack traffic; and HTTP GET Flood bombards web applications with requests that appear to be legitimate.

An ICMP attack exploits the protocol's functionality to overwhelm a target system. ICMP is primarily used for network diagnostics and error reporting, helping devices communicate issues like unreachable destinations or packet loss. However, attackers manipulate its normal use to cause denial-of-service (DoS) conditions [17]. In an ICMP Flood Attack (Ping Flood), an attacker sends an excessive number of ICMP Echo Requests (ping requests) to a victim system, forcing it to respond to each request with Echo Replies.

This continuous exchange consumes network bandwidth, CPU processing power, and system memory, ultimately disrupting normal operations. The attack's effectiveness is heightened when botnets or

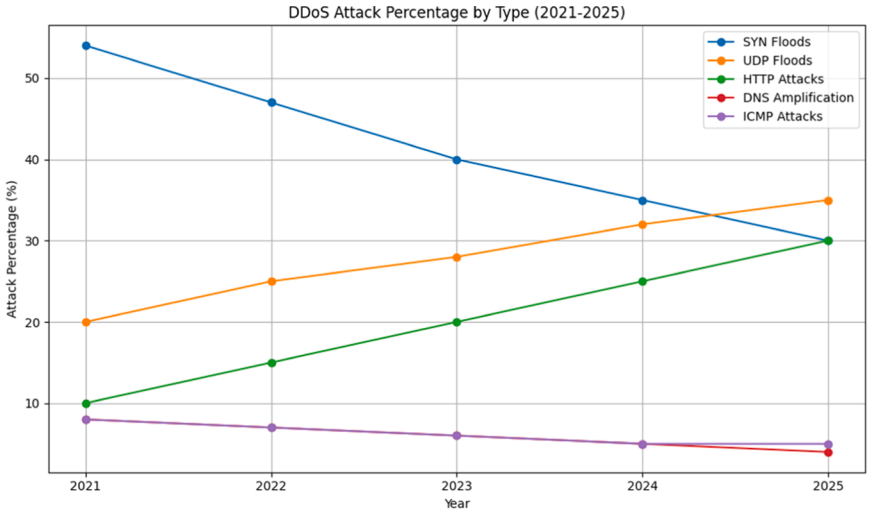


Fig. 1. Trends in DDoS attack types from 2021 to 2025, showing the changing prevalence of SYN floods, UDP floods, HTTP attacks, DNS amplification, and ICMP attacks as a percentage of total attacks. Cloud flare, Inc. (2025). DDoS threat reports. Cloud flare's Blog [16].

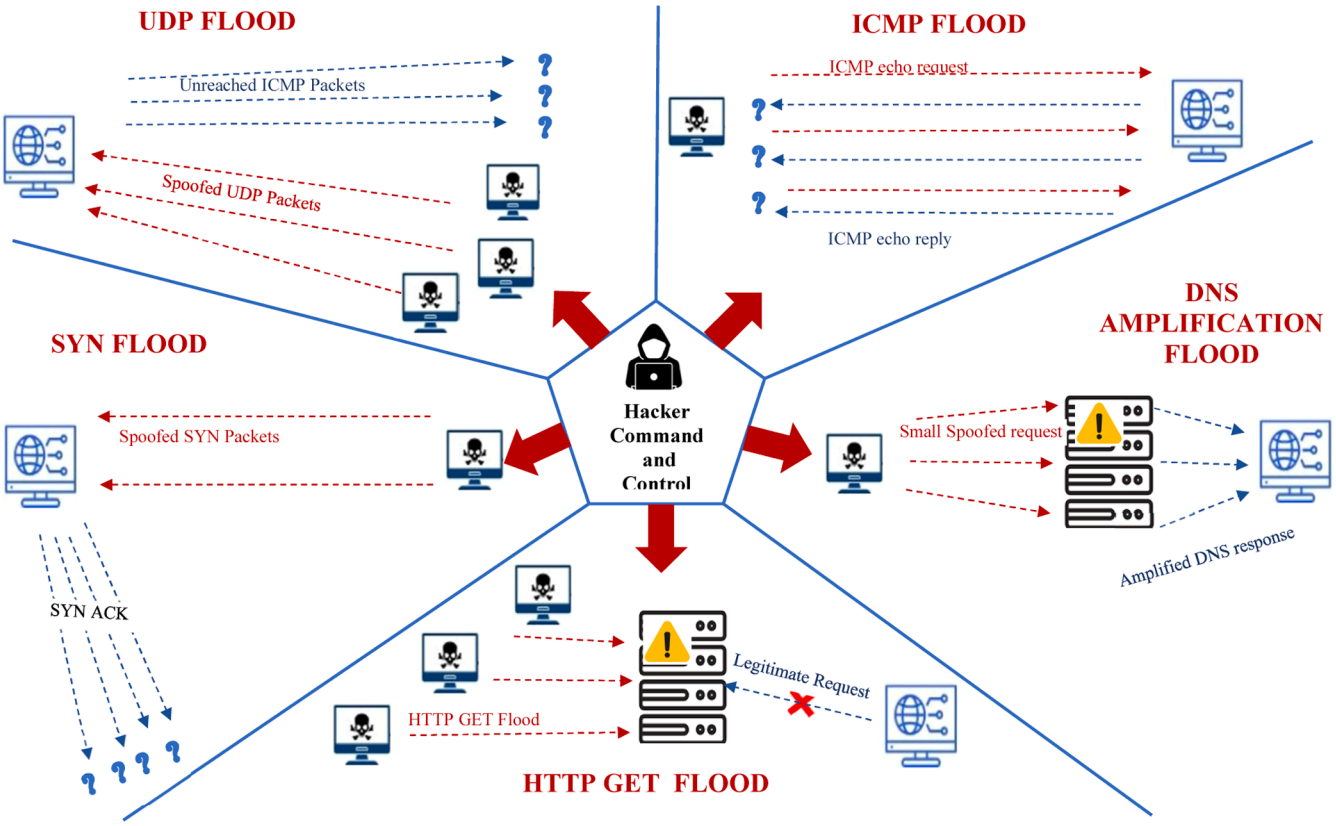


Fig. 2. Illustration of DDoS flooding attack such as (UDP, ICMP, SYN, DNS amplification, HTTP Get) floods.

spoofed IP addresses are used to increase traffic volume, making mitigation difficult. Because the attack method revolves around ICMP packet abuse, it is classified as an ICMP-based attack under the broader category of DDoS (Distributed Denial-of-Service) threats. Organizations mitigate such threats using rate limiting, firewall filtering, and intrusion detection systems (IDS) to restrict excessive ICMP traffic and maintain network availability.

$$R_{ICMP} = \sum_{i=1}^{N} P_{ICMP}(t) \quad (1)$$

Where, R_{ICMP} is the Total ICMP request rate, $P_{ICMP}(t)$ represents the number of ICMP packets sent at time t, N is the number of attacking nodes

A SYN packet is a fundamental part of the TCP handshake procedure that sets up links among endpoints in a network. A flood attack involving SYN overwhelms the TCP handshake process to overwhelm an intended system Typically, a client sends a SYN request to the server in order to establish a connection. The server then responds with a SYN-ACK, anticipating a final ACK to complete the handshake. A SYN flood attack occurs when the attacker sends a large number of SYN requests but never finishes the handshake, leaving the server with connections

that are only partially established. This exhausts system resources, preventing legitimate users from accessing the service. Since DDoS attacks involve multiple compromised devices (botnets) sending these requests simultaneously, the impact is amplified, making mitigation difficult. As a result, the server becomes unresponsive, leading to a Denial-of-Service condition that disrupts normal operations [18].

$$R_{SYN} = \sum_{i=1}^{N} P_{SYN}(t) - A_{ACK}(t) \quad (2)$$

Where, R_{SYN} is the unanswered SYN requests, $P_{SYN}(t)$ represents the SYN packets sent at time t, $A_{ACK}(t)$ is the ACK (acknowledgment) responses received.

UDP is a connectionless transport protocol used for fast, lightweight communication without requiring a handshake process like TCP. While efficient for applications like video streaming and VoIP, its lack of built-in security makes it vulnerable to DDoS. A UDP flood attack exploits this by overwhelming a target system with a massive number of UDP packets sent to random ports. The system attempts to process each packet and respond, but since the requests are often spoofed or invalid, it ends up sending ICMP Destination Unreachable messages, consuming bandwidth and system resources. When executed at scale using botnets, this attack can saturate network capacity, making legitimate services inaccessible [19].

$$R_{UDP} = \sum_{i=1}^{N} P_{UDP}(t) * \mu_{UDP} \quad (3)$$

Where, R_{UDP} is the total UDP traffic rate, $P_{UDP}(t)$ represents the number of UDP packets sent at time t. μ_{UDP} is the average UDP packet size.

A key technology called DNS converts domain names that are understandable by humans into IP addresses so that users can navigate the internet with ease. However, as a form of DDoS attack, assailants use open DNS resolvers to initiate DNS amplification attacks. Using a spoof source IP address (the victim's IP), the attacker sends brief DNS queries to several DNS servers in this attack. These servers unintentionally overload the victim's network with excessive traffic by responding with massive amplified replies. Since the amplification ratio can be significant, even a small request can generate a massive response, overload the target system and make legitimate services inaccessible [20].

DNS amplification exploits spoofed requests to generate disproportionately large responses, where the Amplified DNS traffic rate R_{DNS} is calculated as the product of the number of DNS queries sent $P_{DNS}(t)$ and the Amplification factor α_{DNS} which represents the ratio of response size to request size. DNS amplification uses spoofed requests to generate large responses:

$$R_{DNS} = \sum_{i=1}^{N} P_{DNS}(t) * \alpha_{DNS} \quad (4)$$

HTTP is the foundation of web communication, enabling browsers and applications to request and retrieve content from servers. However, attackers exploit this protocol in HTTP flood attacks, a kind of DDoS attack that overwhelms a web server with excessive HTTP GET or POST requests. Unlike traditional volumetric attacks, HTTP floods operate at Layer 7 (Application Layer), making them harder to detect since the traffic appears legitimate. Attackers often use botnets or compromised devices to generate a massive number of requests, exhausting the server's processing power and bandwidth, ultimately causing service disruptions [21].

HTTP GET flood overwhelms a web server with excessive requests:

$$R_{HTTP} = \sum_{i=1}^{N} P_{HTTP}(t) * L_{HTTP} \quad (5)$$

Where, R_{HTTP} is the Total HTTP request rate, $P_{HTTP}(t)$ represents the number of GET requests sent. L_{HTTP} is the Load impact per request. When a network is overwhelmed, packet loss increases, The percentage of lost packets (L) is determined by the ratio of attack traffic volume (T_{attack})to the total network traffic (T_{total})

$$L = \frac{T_{attack}}{T_{total}} * 100 \quad (6)$$

4.2. Mapping critical points of DDoS flooding attacks in 5G network

Fig. 3 provides a comprehensive mapping of the 5G network, highlighting critical points vulnerable to DDoS flooding attacks, which emphasizing key elements, the manner in which are connected, and potential security risks. Prior to connecting to the core network, the User Equipment (UE) communicates with the Next Generation Radio Access Network (NG RAN). A number of crucial tasks are involved in the core network: Device access and mobility are managed by the Access and Mobility Management Function (AMF), user data flow is facilitated by the User Plane Function (UPF), and session continuity is managed by the Session Management Function (SMF). The Network Data Analytic Function (NWDAF) offers network insights, the Policy Control Function (PCF) enforces user policies, the Network Slice Selection Function (NSSF) facilitates slice selection, the Authentication Server Function (AusF) guarantees secure authentication, and the Unified Data Management (UDM) & User Data Repository (UDR) store and retrieve subscriber data.

Interfaces that allow communication between these components are also shown in the diagram, including N1 (UE-AMF), N2 (NG RAN-AMF), N3 (NG RAN-UPF), N4 (SMF-UPF), and N6 (UPF-Data Network). DDoS Flooding threats, such as UDP Flood, ICMP Flood, DNS Amplification Flood, SYN Flood, and HTTP GET Flood, have also been identified as potential attack vectors that target different parts of the network infrastructure were mapped in the 5G design [22]. DDoS Flooding threats have also been identified as possible attack vectors that target several network infrastructure components that were mapped in the 5G architecture. These threats include UDP Flood, ICMP Flood, DNS Amplification Flood, SYN Flood, and HTTP GET Flood.

4.2.1. Quality of service (QoS) based identification and detection of 5G flooding attacks

Monitoring QoS measurements in 5G networks is critical for detecting flooding assaults, which impair service. Bit rate indicators, transmission rates, and error rates can help detect anomalous traffic surges caused by volumetric DDoS attacks. Signal quality measures indicate interference or attack-induced congestion User and device identification attributes help to track affected areas and assault trends. The correct connection of these factors provides effective identification and mitigation of 5G-based flooding concerns [23].Bit rate indicators such as dl_bitrate and ul_bitrate help monitor download and upload rates, and sudden increases may indicate flooding attacks. Transmission rates (dl_tx, ul_tx) and total bytes sent (bearer_0_dl_total_bytes, bearer_0_ul_total_bytes) can detect excessive traffic, indicating a volumetric ddos attack. While decoder performance (turbo_decoder_avg, turbo_decoder_max, and turbo_decoder_min) helps detect network stress, error and retransmission rates (ul_retx, ul_err, and dl_err) may reveal attack-induced congestion. Poor communication due to interference or flooding attacks can be indicated by signal quality indicators (cqi, epre, and pusch_snr). While unique identifiers like 5 g_tmsi, imeisv, and rnti help monitor attack trends, user and device identification attributes like amf_ue_id, ran_ue_id, tac, cell_id, and ran_plmn are essential for identifying impacted individuals and locations[24]. Attack detection metrics, such as attack, malicious, and attack_number, offer information on known threats, whereas registration status (registered) indicates if a device is legitimately connected or faked. QoS and latency measures, such as dl_mcs, ul_mcs, ul_phr, ul_rank, and initial_ta, can detect unexpected shifts in modulation schemes, power headroom, and timing advance, which may indicate network abuse. Volumetric attacks have an

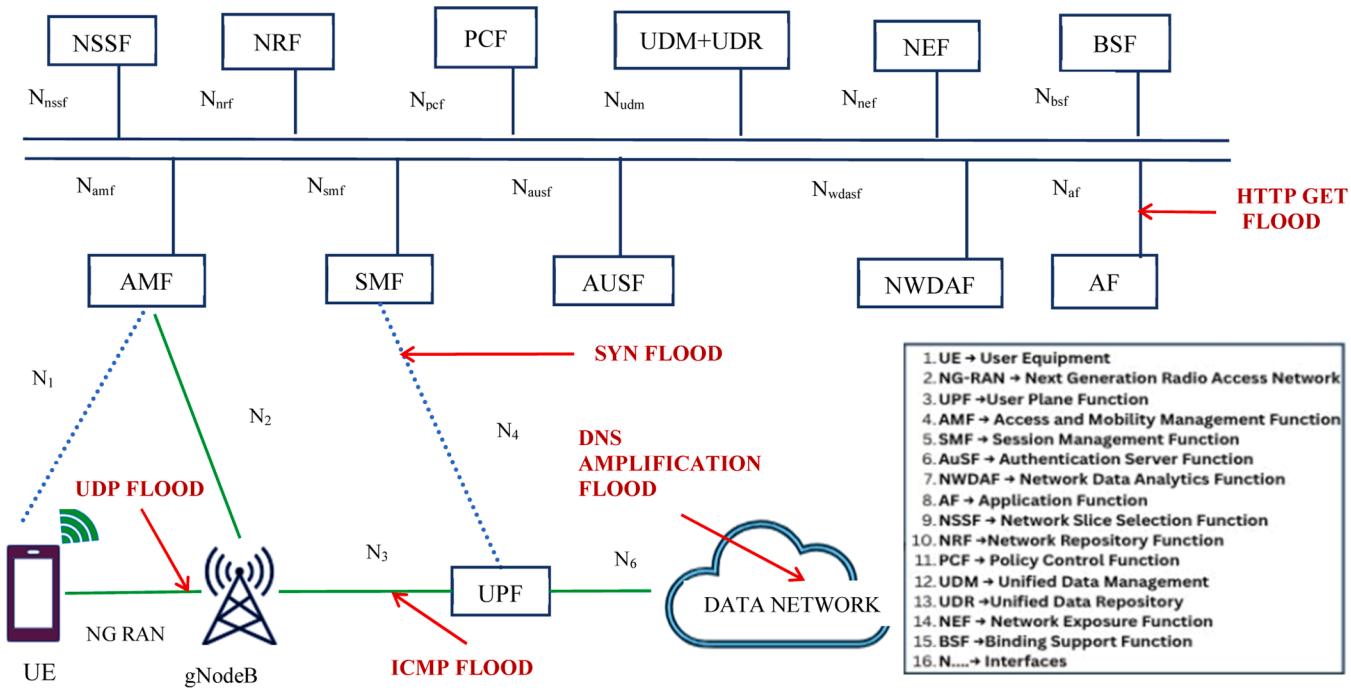


Fig. 3. Illustrates the Mapping of 5 G network architecture and point of DDoS flooding attacks.

excessive bitrate, transmission rates, and total bytes; protocol exploits have aberrant error rates and retransmissions; and resource depletion attacks have decreased CQI (Channel Quality Indicator) and SNR (Signal-to-Noise Ratio) and modulation schemes. Proper correlation of these features enables the effective detection and mitigation of 5G-based flooding assaults [25].

5. Related works

Federated learning is being used more and more in recent DDoS detection research to solve scaling and security in distributed networks. Industrial IoT-focused studies in 2021 achieved 95.4 % accuracy using FL but face scalability challenges in 5G's dense device ecosystems [26]. Ahmetoğlu and Das present a comprehensive survey of cyber-attack detection methods, datasets, and classification strategies, emphasizing both traditional and deep learning techniques. They highlight the limitations of existing datasets and advocate for more realistic, dynamic benchmarks to address evolving threat landscapes [27].Further investigations in 2022 introduced hybrid models that integrate Convolutional Neural Networks (CNNs) with FL, achieving an approximately 98 % accuracy rate while ensuring data privacy [28]. This combination leverages CNNs' capability to extract spatial features from network traffic, thereby enhancing precision in attack detection. By decentralizing model training across multiple edge devices, FL significantly reduces centralized data exposure, mitigating privacy risks in IoT networks.

Traditional machine learning demonstrates high accuracy in DDoS attack detection, yet its reliance on centralized data processing raises privacy concerns for distributed network environments [29]. This limitation motivates the exploration of federated learning and ensemble methods for more adaptive and privacy-preserving security solutions

Adaptive Federated Learning (FLAD) frameworks have shown promising results in dynamic threat detection, achieving 96.8 % accuracy by emphasizing devices with complex attack patterns. These methods modify aggregation weights dynamically during training, enhancing resistance against evolving cyber threats [30]. However, FLAD has not been evaluated in 5G-specific contexts, such as network slicing and ultra-reliable low-latency communication (URLLC), where diverse traffic patterns and stringent Quality of Service (QoS) requirements necessitate specialized optimization.

Similarly, SDN-integrated Federated Learning models have demonstrated 97.1 % accuracy in controlled environments, highlighting their scalability within software-defined networks (SDNs) through centralized management [31]. Although effective for large-scale traffic analysis, these frameworks require further validation in real-world 5G deployments, particularly in distributed control architectures, where edge nodes and core networks must synchronize to respond effectively to cyber threats.

Further research has been conducted on Ensemble Federated Learning (EFL) to enhance data privacy while reducing computational costs. Wang et al. introduced a stacking ensemble GRU optimization algorithm within a federated learning framework for electricity theft detection in smart grids, achieving high detection rates against cyber-attacks [32]. Shen et al. addressed intrusion detection in heterogeneous IoT networks using ensemble knowledge distillation-based FL, enhancing detection in diverse device environments [33]. Arroyo presented an autoencoder ensemble system using FL for malware detection on IoT devices, highlighting the benefits of unsupervised learning in distributed settings [34].

Existing Federated Learning (FL) frameworks lack latency-aware aggregation for 5G's ultra-reliable low-latency communication (URLLC), limiting their effectiveness in real-time applications. Additionally, no studies have combined FL with ensemble methods, such as stacking and boosting, to address model bias in heterogeneous 5G traffic. Jeffrey et al. applied ensemble learning for anomaly detection in cyber physical systems, demonstrating improved robustness against sophisticated attacks [35]. Ullah et al. combined blockchain and FL for secure intrusion detection in the Internet of Vehicles, ensuring data integrity and privacy [36].Bukhari et al. proposed an asynchronous FL approach with a deep hybrid model to enhance detection performance in Edge IIoT networks while reducing communication overhead [37]. Integrating deep learning, blockchain, bioinspired, and edge-based models enhances intrusion detection in cloud networks by improving accuracy, scalability, and QoS while reducing delay and complexity [38].The integration of AI and blockchain in smart grid systems not only optimizes energy management but also strengthens security by enabling

real-time detection and verification of anomalies or cyberattacks through transparent, tamper-proof data handling [39].

The summary of the related work based on anamoly and DDoS detection with key result values, limitations are presented in Table 1.

Although the subject has improved as a result of these efforts, there are still a number of research gaps. The processing and communication limitations of edge devices are not adequately addressed by many methods, particularly in the context of 5G networks where high throughput and low latency are essential. Although the combination of optimization and ensemble learning approaches shows promise, further study is required to guarantee scalability, flexibility, and effectiveness in extremely resource-constrained and dynamic settings.

6. Proposed work

6.1. System workflow of ensemble federated learning for detection of DDoS attacks

Fig. 4 illustrates the proposed system workflow for Hybrid Ensemble Federated Learning, integrated with SMOTE-Tomek to ensure balanced dataset training. This approach enhances the model's efficiency in detecting Distributed Denial-of-Service (DDoS) attacks, while maintaining privacy-preserving, scalable, and resource-efficient operations tailored for constrained Internet of Things (IoT) edge devices within 5G networks.

The workflow begins with the acquisition of network traffic data from a dedicated DDoS dataset such as NCSRD-DS 5GDDOS, ensuring that the data is representative of real-world attack scenarios in next-generation networks. This raw data undergoes a comprehensive preprocessing phase, which includes cleaning to remove noise and inconsistencies, normalization to standardize feature scales, and feature extraction to derive meaningful attributes from the traffic records. To further enhance computational efficiency and model accuracy, feature engineering is performed using Recursive Feature Elimination (RFE) to systematically identify which is especially important for optimizing performance on resource-constrained edge devices. Also to reduce dimensionality Principal Component Analysis is done. Addressing the common issue of class imbalance in intrusion detection, advanced data sampling techniques such as SMOTE -Tomek Links are applied to create a more balanced dataset, ensuring that minority attack classes are sufficiently represented and improving the robustness of the detection models. Once preprocessed, the dataset is split into training and testing subsets, 80 % of the dataset is for the train split and 20 % of the dataset the test split. The core of the architecture is the Hybrid Ensemble federated learning framework, where multiple edge nodes (e.g., Edge1 to Edge3) each deploy distinct lightweight machine learning models, such as XGBoost, Random Forest, and Gradient Boosting. Each node trains its model locally on its own data partition, maintaining data privacy by sharing only encrypted class probability outputs with a central server rather than raw data for aggregation.

The central server follows to two powerful aggregations for ensemble the class probability predictions. Soft voting and stacking Meta-learner are used for the ensemble the class probability prediction in the aggregation. This aggregation process not only preserves privacy and minimizes communication overhead

Ultimately, this detailed architecture supports real-time, distributed DDoS detection at the edge, tailored for the unique challenges and constraints of 5G-enabled IoT environments.

6.2. Hybrid ensemble federated learning (EFL)Architecture

Fig. 5 illustrates the architecture of hybrid ensemble detection framework grounded in federated learning principles, where three distinct models, XGBoost, Gradient Boosting, and Random Forest are deployed on simulated clients with localized, balanced data partitions for detecting the DDoS flooding attacks classification and benignThe class distribution was updated once over two rounds Also it uses two ensemble strategies for final prediction of classes. Soft voting and another ensemble technique is Stacking Meta learner. The class probability from the each models are as follows

XGBoost computes raw scores (logits) $f_k(x)$ for each class k ϵ {0,1,2,3,4,5} by summing the outputs of all trees assigned to that class. These scores are converted to probabilities using the softmax function

Table 1
Summary of recent federated learning-based anomaly and DDoS detection approaches.

Paper (Authors, Year)	Objective/ Main Contribution	Technique/ Framework	Evaluation Metrics	Key Results	Limitations / Research Gap
Li et al. (2021)[26]	Mitigate DDoS in IIoT using FL	Federated Learning Architecture	Accuracy: 95.4 %,	Privacy-preserving detection in IIoT	Limited testing in 5G or edge networks
Lv et al. (2022) [28]	DDoS detection using CNN feature extraction in FL	CNN + Federated Learning	Accuracy: ~98 %, Precision: ~96 %, Recall: ~97.8 %, F1-score: ~94 %	Effective CNN-based feature learning in FL	Lacks real-time optimization for 5G latency
Tymoshchuk et al. (2024) [29]	Detect/classify DDoS flooding attacks	ML classification (SVM, RF)	Accuracy: 99.35 %, Precision: >99 %, Recall: >99 %, F1-score: >99 %	High accuracy in attack classification	Centralized training (no FL integration)
Doriguzzi-Corin & Siracusa (2024) [30]	Adaptive FL for dynamic DDoS detection	Adaptive Federated Learning (FLAD)	Accuracy: 96.8 %,	Dynamic model adaptation improves detection	5G-specific network dynamics not addressed
Fotse et al. (2024) [31]	FL-based DDoS detection in SDN	Federated Learning + SDN	Accuracy: 97.1 %,	Scalable detection in SDN environments	Requires real-world 5G validation
Wang et al. (2024) [32]	Detect theft via stacking ensemble GRU in FL	Stacking ensemble GRU + FL	CIC-IOT2023:Accuracy: 97.47 %Precision: 97.51 %Recall: 97.44 %F1-score: 97.47 %CIC-IDS2018: Accuracy: 94.55 %Precision: 94.64 %Recall: 94.52 %F1-score: 94.55 %	High detection with CIC-IOT2023	Edge/5G optimization not addressed
Shen et al. (2024) [33]	Intrusion detection via ensemble knowledge distillation FL	Ensemble KD-based FL	F1-score: 98.3 %Recall: 97.8 %	Enhanced detection	Scalability for 5G/ edge untested
Arroyo (2024) [34]	FL-based autoencoder ensemble for malware	Autoencoder Ensemble FL	F1-score: 97.9 %AUROC: 98.1 %	Effective distributed detection	Edge/5G adaptation not tested
Jeffrey et al. (2024) [35]	Ensemble learning for anomaly detection	Ensemble learning	Accuracy: 96.8 %F1-score: 96.4 %	Robust detection	No FL/edge/5G focus
Ullah et al. (2025) [36]	Blockchain-based FL for secure detection	Blockchain + FL	Accuracy: 97.8 %F1-score: 97.7 %	Enhanced privacy/ security	Overhead/latency for 5G not studied
Bukhari et al. (2024) [37]	Asynchronous FL with deep hybrid model	Asynchronous FL + Hybrid	Accuracy: 98.7 %F1-score: 98.6 %	Handles asynchronous, heterogeneity	Optimization for 5G edge devices needed

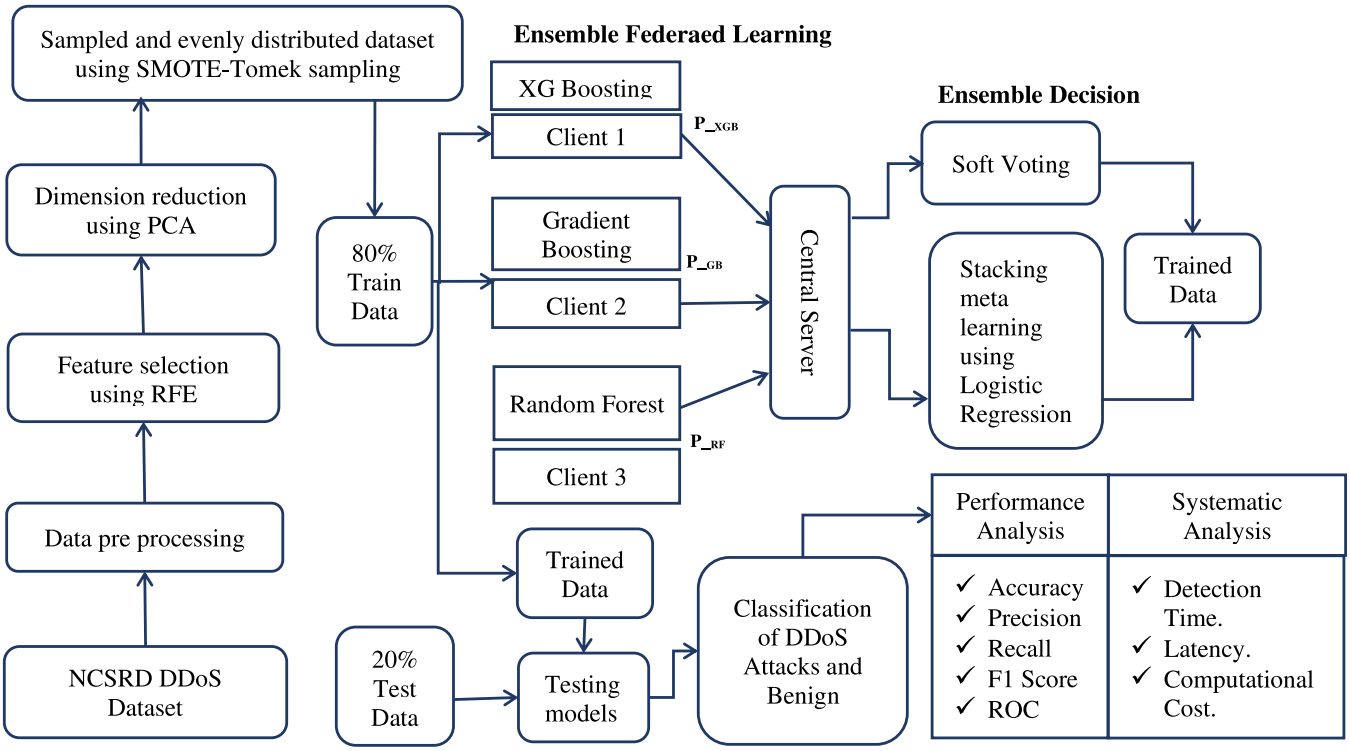


Fig. 4. Proposed system architecture of hybrid ensemble federated learning principles using SMOTE-Tomek for efficient DDoS detection.

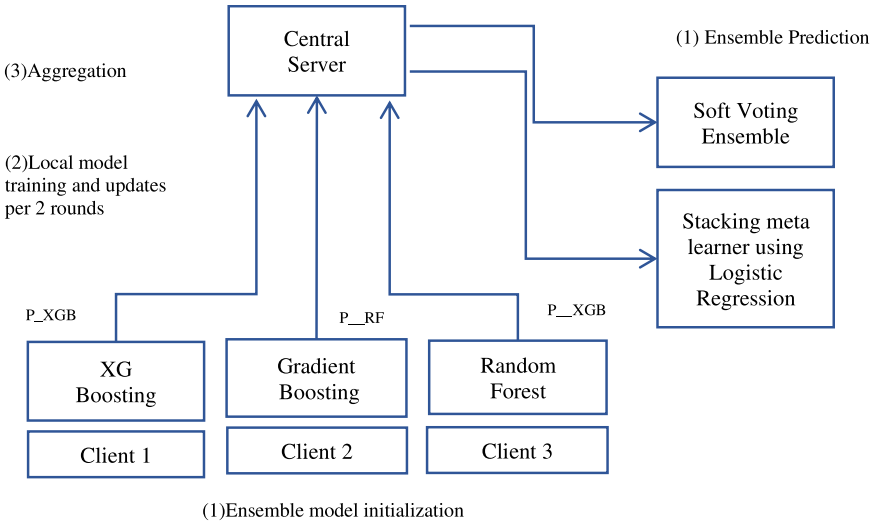


Fig. 5. Architecture of hybrid ensemble federated learning (XGB, GB, RF) with soft voting and stacking Meta-learner.

$$P_{XGB}(y=k|x) = \frac{e^{(f_k\ (x))}}{\sum_{j=0}^{5} e^{(f_j\ (x))}} \tag{7}$$

Where $f_k\ (x)$ is the raw score for class k (sum of tree outputs for class k). P_{XGB} is sent as the class probability for sample x. f_j(x) denotes the sum of the outputs of all decision trees in the ensemble that correspond to class j.

Gradient Boosting (GB):

Gradient Boosting also computes a raw score f_k(x) for each class k ϵ {0,1,2,3,4,5} . The class probabilities are then obtained using the soft-max function, which ensures the output is a valid probability distribution across all classes.

$$P_{GB}(y=k|x) = \frac{e^{(f_k\ (x))}}{\sum_{j=0}^{5} e^{(f_j\ (x))}} \tag{8}$$

Random Forest

Random Forest computes the class probabilities by aggregating the individual predictions from each decision tree in the ensemble. For a given input x, each tree casts a vote for a class label, and the final class probability distribution is obtained by averaging these votes across all trees, ensuring a valid probabilistic output over all classes.

$$P_{RF}(y=k|x) = \frac{1}{T}\sum_{t=1}^{T}[f_t\ (x)=k] \tag{9}$$

Where: T is the total number of trees in the forest. $f_t(x)$ is the

predicted class label for sample x by the tree t.

$[f_t(x) = k]$ is the indicator function, which is 1 if free t predicts class k, and 0 otherwise. Each tree outputs of a probability vector and the final probability is the average of these vectors across all trees.

Soft voting Ensemble:

Soft voting ensemble predicts the class probability distribution, where each base classifier sends the class probability to the central server for each test sample. The final prediction is made by the averaging these class probabilities and selecting the class with highest average probability.

Step1: Model Training

Train each base model(XGB, GB, RF) Independently on its respective data partition based on the DDoS Flooding attack classification on the benign.

Step 2: Probability prediction

For each test sample x each model 'i' outputs a probability vector

$$\widehat{P}_i(y = c \mid x),\ c \in \{0,\ 1,\ 2, 3, 4, 5\} \tag{10}$$

This is the predicted probability that sample x belongs to class c according to model i.

Step 3: Aggregation

For each class C, compute the average probability across all n models.

$$\overline{P}(y = c \mid x) = \frac{1}{n}\sum_{i=1}^{n} \widehat{P}_i(y = c \mid x) \tag{11}$$

Step 4: Final Prediction

Assign the class with the highest average probability as the predicted label

$$\widehat{y}(x) = \underset{c}{argmax}\overline{P}(y = c \mid x) \tag{12}$$

Soft Voting leverages the complementary strengths of multiple models, often resulting in improved accuracy and better generalization compared to relying on any single model.

Stacking Meta Learner:

Stacking is an advanced ensemble technique where the outputs (class probabilities) of several base models are used as input features for a second level model (meta-learner), which learns to optimally combine them.

Step 1: Model Training:

Train each base model (XGB, GB, RF) independently on its local data.

Step 2: Probability collections:

For each test sample x, collect the probability vectors from all base models

$$stacked\ Input\ (x) = \ \widehat{P}_0(y = c \mid x), \widehat{P}_1(y = c \mid x),\ \widehat{P}_2(y = c \mid x) \tag{13}$$

For three models and six classes, this yields an 18 dimensional feature vector per sample.

Step 3: Meta –Learner training

Accumulate stacked input vectors and corresponding true labels (from a validation set or via cross validation train a meta learner here logistic regression is used on these features to learn the mapping.

$$\widehat{y}_{meta}(x) = MetaModel(stacked\ Input\ (x)) \tag{14}$$

Step 4: Final Prediction:

For each test sample, the meta- learner outputs a probability vector

$$\widehat{P}_{meta}(y = c \mid x) \tag{15}$$

Assign the class with the highest meta-learner probability

$$\widehat{y}_{final}(x) = \ \underset{c}{argmax}\widehat{p}_{meta}(y = c \mid x) \tag{16}$$

Stacking allows the meta learner to discover comples patterns and relationships among base models outputs, often achieving superior performance and generalization compared to soft voting or any other individual.

6.3. Data preparation for proposed hybrid ensemble federated learning to detect DDoS attacks

The Data Preparation and data balancing Process for Ensemble Federated Learning, outlining the sequential steps required to refine the dataset before model training. It begins with the NCSRD - DS-5 G DDoS dataset, which undergoes data preprocessing to handle missing values, scale features, and encode categorical variables [40]. Next, SMOTE-Tomek sampling is applied to balance the dataset, followed by Recursive Feature Elimination (RFE) for selecting the most critical features.

NCSRD-DS-5GDDoS Dataset

Utilizing the NCSRD-DS-5GDDoS dataset, the framework records attack traffic logs from constrained IoT edge devices, including CPE Boxes, Wi-Fi 6 Mobile Hotspots, Industrial 5 G Routers, and Raspberry Pi 4 [38]. The shape of the dataset is: (614,428, 58).

Columns: ['_time', 'imeisv', '5 g_tmsi', 'amf_ue_id', 'bearer_0_apn', 'bearer_0_dl_total_bytes', 'bearer_0_ip', 'bearer_0_ipv6′, 'bearer_0_pdu_session_id', 'bearer_0_qos_flow_id', 'bearer_0_sst', 'bearer_0_ul_total_bytes', 'bearer_1_apn', 'bearer_1_dl_total_bytes', 'bearer_1_ip', 'bearer_1_pdu_session_id', 'bearer_1_qos_flow_id', 'bearer_1_sst', 'bearer_1_ul_total_bytes', 'dl_bitrate', 'ran_id', 'ran_plmn', 'ran_ue_id', 'registered', 'rnti', 't3512′, 'tac', 'tac_plmn', 'ue_aggregate_max_bitrate_dl', 'ue_aggregate_max_bitrate_ul', 'ul_bitrate', 'bearer_1_ipv6′, 'cell', 'ul_retx', 'ul_err', 'ul_mcs', 'ul_n_layer', 'ul_path_loss', 'ul_phr', 'ul_rank', 'dl_err', 'dl_mcs', 'dl_retx', 'dl_tx', 'cqi', 'epre', 'initial_ta', 'p_ue', 'pusch_snr', 'ri', 'turbo_decoder_avg', 'turbo_decoder_max', 'turbo_decoder_min', 'ul_tx', 'cell_id', 'attack', 'malicious', 'attack_number']

6.4. Balancing the imbalanced dataset using SMOTE Tomek links

Data sampling is essential for research on constrained edge devices, as implementing a large dataset presents significant challenges. It is essential to balance the dataset for detecting the attacks with high performance. To balance the dataset, SMOTE-Tomek sampling is employed, balancing the dataset by generating synthetic samples for the minority class while refining decision boundaries through the removal of overlapping samples. This approach enhances model accuracy and generalization without compromising detection efficiency. Fig. 6 illustrates the Flowchart representation of SMOTE-Tomek Integration for Class Balancing in DDoS Detection on Constrained Edge Devices.

SMOTE (Synthetic Minority Oversampling Technique)

Datasets, benign traffic often overwhelms attack samples, leading to biased model predictions. With a 94 % benign vs. 6 % attack distribution, classifiers struggle to detect rare attack patterns. Dataset of 80,000 samples is taken for training the edge devices. However, with only 6000 samples per attack type, the available data falls short of the necessary threshold for reliable classification. To address this limitation, synthetic data generation is implemented to enhance sample diversity. SMOTE-Tomek sampling is utilized to correct class imbalance by generating additional attack samples while refining decision boundaries through the removal of overlapping instances. This approach ensures feature diversity preservation, strengthens model generalization, and optimizes attack detection performance without introducing over fitting. With a final dataset composition of 13,334 instances for each classification, the required sample size for effective classification is met. This structured distribution guarantees adequate representation of both benign and attack classes, leading to improved detection accuracy and well-balanced model training. For testing 1200 instances from each classification are used.

SMOTE generates synthetic samples for minority classes using k-nearest neighbours (KNN) interpolation, ensuring balanced class representation without duplicating existing data.

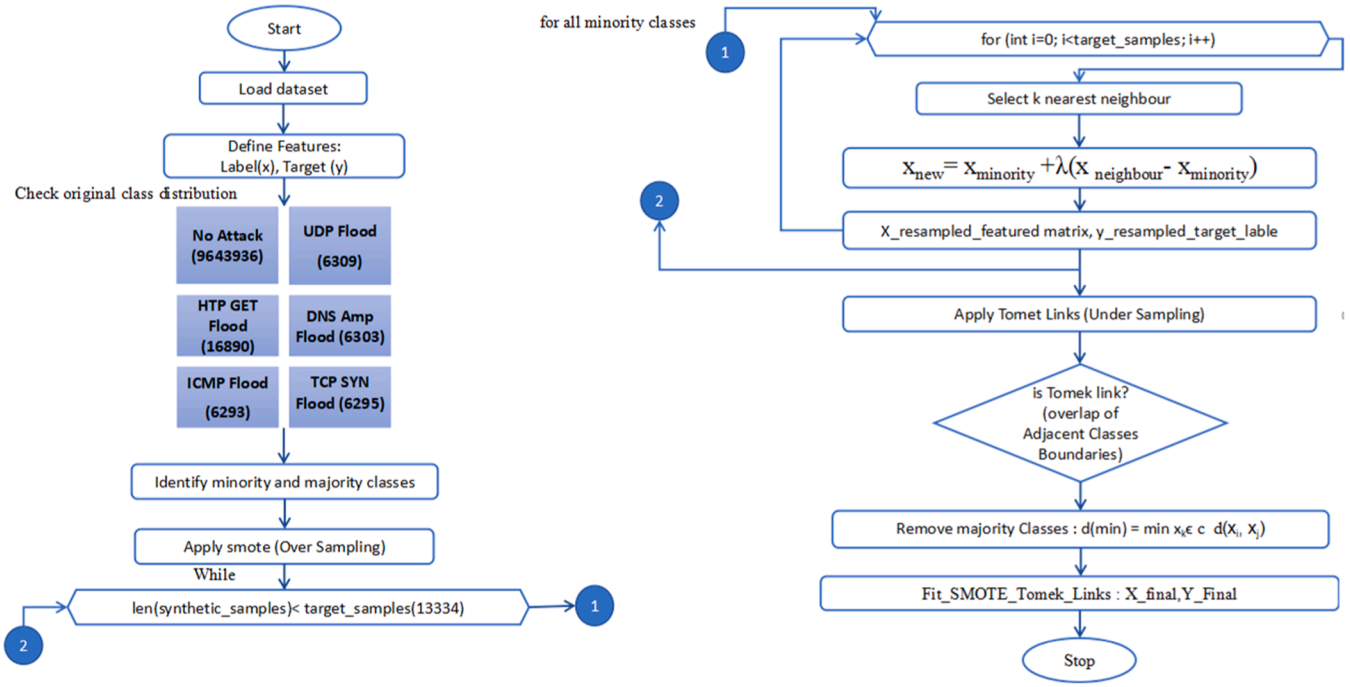


Fig. 6. Flowchart representation of SMOTE-Tomek integration for class balancing in DDoS detection on constrained edge devices.

$$x_{new} = x_{minority} + \lambda\left(x_{neighbour} - x_{minority}\right) \tag{14a}$$

Where x, represents the data point in the features space. Specifically, $x_{minority}$is the sample from the minority class (underrepresented attack type). $x_{neighbour}$ is the selected nearest neighbour from the same minority class. x_{new} is the synthetically generated sample, λ is the random interpolation factor in (0,1)(0,1).

The Tomek Links method refines decision boundaries by systematically eliminating overlapping majority-class samples. First, the algorithm identifies closest pairs, where a majority-class sample x_i is nearest to a minority-class sample x_j. Next, class separation is analysed if x_i is closer to x_j than to any other sample within its own class, it indicates class overlap, which can negatively impact classification accuracy. Finally, such overlapping majority samples are removed, reducing misclassification risks and ensuring a clearer boundary between minority and majority classes. This process enhances the effectiveness of SMOTE-Tomek sampling, improving generalization and model performance.

$$d_{(min)} = \min_{x_k \in \mathbb{C}} d\left(x_i, x_j\right) \tag{15a}$$

Where, x_i represents a sample from the majority class. x_j, represents a sample from the minority class, $d\left(x_i, x_j\right)$, is the Euclidean distance (or other distance metric) between $x_i$andx_j, x_k, represents a neighbouring sample from the same majority class, C is the set of all samples from the dataset.

6.5. Feature selection using recursive feature elimination (RFE)

The feature selection is an important process to retain only the most essential attributes while removing irrelevant or redundant data. This process is crucial for reducing dataset size, enhancing computational efficiency, and simplifying complexity without compromising model performance. Recursive Feature Elimination (RFE) systematically ranks features based on their importance and eliminates the least influential ones, ensuring that only the most relevant attributes contribute to model training. The formula used in RFE is:

$$w_j = \frac{|\beta_j|}{\sum |\beta_k|} \tag{16a}$$

Where, w_j is weight assigned to feature j, β_j represents model coefficient for feature j, $\sum |\beta_k|$ represents sum of absolute coefficients for all features.

This formula helps determine the significance of each feature, ensuring that the dataset is refined to preserve only high-impact variables. After applying RFE, 36 essential features remain, which effectively reduces complexity, improves model efficiency, and enhances classification accuracy.

(101,640, 36)

['dl_bitrate', 'ul_bitrate', 'ul_retx', 'ul_err', 'dl_err', 'dl_tx', 'ul_tx', 'cqi', 'epre', 'pusch_snr', 'amf_ue_id', 'ran_ue_id', 'tac', 'cell_id', 'tac_plmn', 'ran_plmn', 'attack', 'malicious', 'bearer_0_dl_total_bytes', 'bearer_0_ul_total_bytes', 'dl_mcs', 'ul_mcs', 'attack_number', 'registered', 'rnti', '5 g_tmsi', 'imeisv', 'ul_phr', 'ul_rank', 'initial_ta', 'turbo_decoder_avg', 'turbo_decoder_max', 'turbo_decoder_min', 'bearer_0_qos_flow_id', 'bearer_1_qos_flow_id'].

6.6. Dimensionality reduction using PCA principal component analysis

By converting the data into an orthogonal subspace that retains the most significant variance, Principal Component Analysis (PCA) was used to decrease the initial 35-dimensional feature space to 15 components. After standardizing the input features to provide a consistent contribution, the covariance matrix was calculated, and principal components were extracted using eigen decomposition. To capture the most instructive directions in the dataset, the top 15 components—those linked to the biggest eigenvalues—were chosen. In situations where bandwidth and local resources are limited, such as DDoS attack categorical in federated learning, this dimensionality reduction technique is particularly useful since it increases computing efficiency and reduces over fitting while maintaining interpretability. The size and column list of the dataset after pca as follows,

Size of the dataset (rows, columns):

(101,640, 16) The dataset contains columns labeled "pca_component_0″ through "pca_component_14," with the target

column named "attack_number." All columns are of object data type. Using the balanced dataset generated through the hybrid pipeline PCA_RFE_SMOTE_Tomek, federated learning is simulated across three edge devices to detect DDoS attacks over a 5G network using ensemble machine learning techniques.

7. Experimental analysis

Implementation and Experimental Setup

All experiments were carried out in Google Colab, a cloud-based Python platform offering GPU acceleration and access to modern data science libraries. The following libraries were employed:

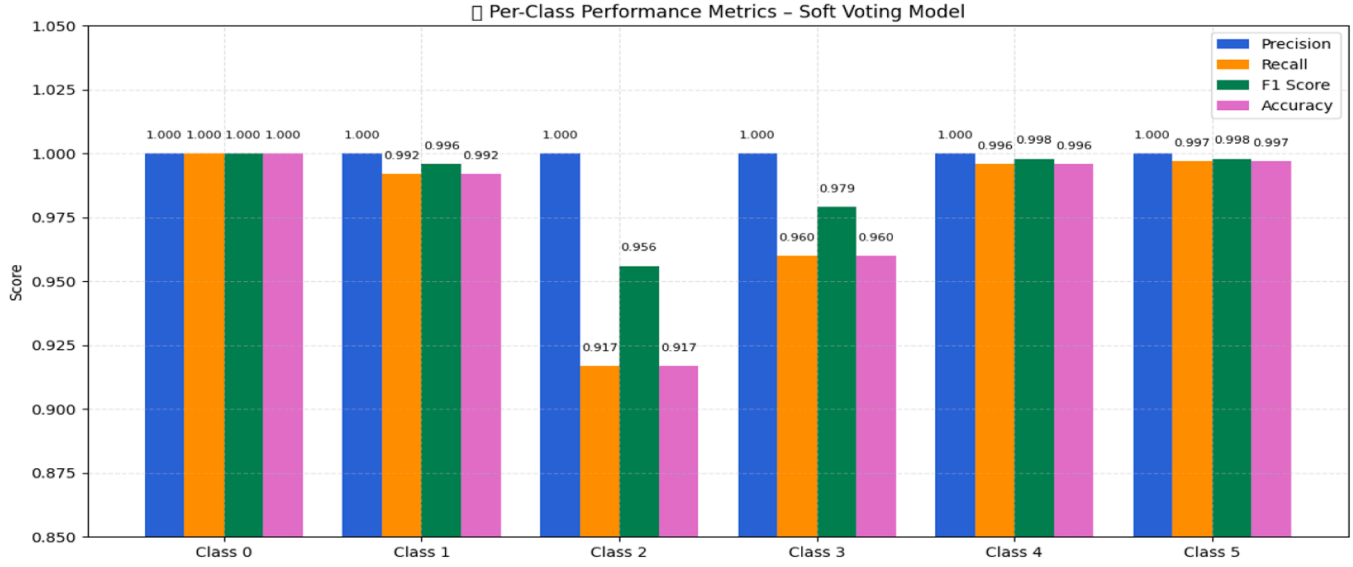
- pandas (v2.2.3) for data handling
- numpy (v1.26.4) for numerical computation
- matplotlib (v3.10.3) for visualization
- scikit-learn (v1.6.1) for model development, preprocessing, and evaluation

To support ensemble modeling, XGBoost (v3.0.1) was used, specifically leveraging the XGBClassifier. Additionally, scikit-learn components such as RandomForestClassifier, GradientBoostingClassifier, StandardScaler, label_binarize, and a range of evaluation metrics (accuracy, precision, recall, F1-score, ROC, and AUC) were integrated into the detection pipeline. Preprocessing steps included data normalization and feature selection, followed by model training and evaluation using train_test_split. The entire workflow from code execution and model training to performance analysis was executed within the Google Colab environment, ensuring reproducibility and streamlined accessibility.

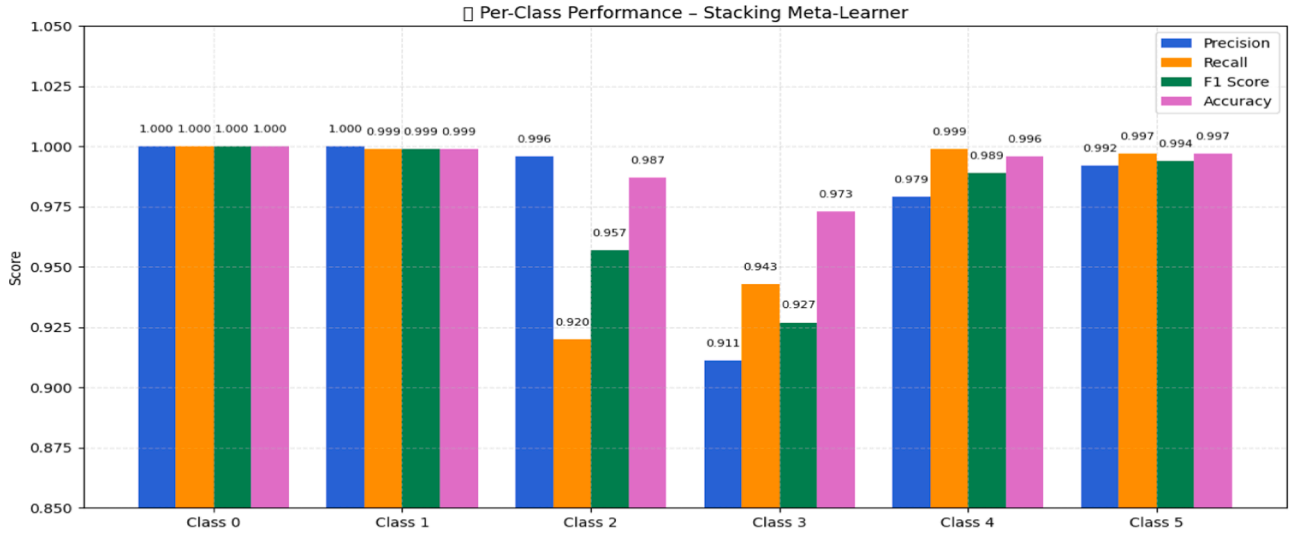
8. Performance evaluation

The proposed ensemble federated learning framework for DDoS detection on constrained IoT edge devices in 5G networks was rigorously evaluated using multiple performance metrics, including accuracy, precision, recall, F1-score, and Area Under the ROC Curve (AUC).

Fig. 7a and 7b Represents the classification performance of a hybrid



(a)



(b)

Fig. 7. Fig. 7a and Fig. 7b represents the classification performance of a hybrid ensemble federated learning model in detecting various types of DDoS flooding attacks, measured across performance metrics Accuracy, Precision, recall, F1 score.

ensemble federated learning model in detecting various types of DDoS Flooding attacks, measured across performance metrics Accuracy, Precision, recall, F1 score. The values are shown in the Table 2. The comparative evaluation of Soft Voting and Stacking Meta-Learner across various DDoS attack types reveals insightful performance trends. For benign traffic (No Attack), both models achieve perfect scores across all metrics, demonstrating their capability to confidently identify safe traffic. In the case of ICMP Flood and TCP SYN Flood, both models remain highly effective, with Stacking slightly edging out Soft Voting on recall and F1 score for ICMP Flood, indicating improved sensitivity. For HTTP GET Flood, Soft Voting exhibits higher precision, while Stacking shows better recall, leading to comparable F1 scores but with Stacking attaining notably higher accuracy. DNS Amplification detection shows a reversal—Soft Voting achieves stronger precision and F1, whereas Stacking demonstrates slightly higher recall and overall accuracy, suggesting a trade-off between false positives and detection completeness. Finally, for UDP Flood, both models perform similarly with excellent scores, although Stacking achieves marginally better recall. Overall, while Soft Voting provides consistently high precision and fewer false positives, Stacking Meta-Learner improves recall and accuracy in some cases, especially for complex attack patterns.

The Fig. 8 displays a side-by-side comparison of confusion matrices from two models—Soft Voting and Stacking Meta-Learner—used for classifying DDoS Flooding Attack types. Both matrices show strong diagonal dominance, indicating high accuracy in correctly predicting most attack categories. The Soft Voting model demonstrates cleaner predictions with minimal off-diagonal values, especially for Class 2 and Class 3, while the Stacking Meta-Learner shows slightly more scattered misclassifications, notably between Class 3 and 4. Despite that, both models excel at identifying "No Attack" and ICMP Flood traffic with near-perfect precision. This visual comparison effectively highlights the models' respective strengths, with Soft Voting favoring consistency and Stacking showcasing deeper learning potential at the cost of occasional confusion among similar attack patterns.

Table 3 presents a comprehensive evaluation of traditional classifiers and ensemble federated learning (EFL) approaches based on key performance metrics, with the corresponding graphical representation illustrated in Fig. 9. Among standalone models, both Random Forest and XGBoost achieve identical and impressive scores, with accuracy, precision, recall, and F1-score at 0.9759 and an AUC of 0.9991, indicating robust and consistent classification. Gradient Boosting trails slightly across all metrics, particularly in AUC at 0.9978. The baseline EFL (per round) performs commendably at 0.9688 across core metrics, suggesting strong generalization. Notably, EFL-Soft Voting enhances this further with an accuracy of 0.9778 and a marginally improved AUC of 0.9992, highlighting the benefits of aggregating predictions. The EFL Stacking model outperforms all, achieving the highest across-the-board scores 0.979 for each core metric and the top AUC of 0.9993 demonstrating its superior ability to synthesize client outputs and distinguish between attack types with exceptional precision.

Table 2
Classification performance of a hybrid ensemble federated learning model (soft voting and stacking) in detecting various types of DDoS flooding attacks, measured across performance metrics.

Class	Attack Type	Model	Precision	Recall	F1 Score	Accuracy
0	No Attack	Soft voting	1.000	1.000	1.000	1.000
		Stacking	1.000	1.000	1.000	1.000
1	ICMP Flood	Soft voting	1.000	0.992	0.996	0.992
		Stacking	1.000	0.999	0.999	0.999
2	HTTP GET Flood	Soft voting	1.000	0.917	0.956	0.917
		Stacking	0.996	0.920	0.957	0.987
3	DNS Amplification	Soft voting	1.000	0.960	0.979	0.960
		Stacking	0.911	0.943	0.927	0.973
4	UDP Flood	Soft voting	1.000	0.996	0.998	0.996
		Stacking	0.979	0.999	0.989	0.996
5	TCP SYN Flood	Soft voting	1.000	0.997	0.998	0.997
		Stacking	0.992	0.997	0.994	0.997

Table 3 presents a comprehensive evaluation of traditional classifiers and ensemble federated learning (EFL) approaches based on key performance metrics.

The Fig. 10 presents a Receiver Operating Characteristic (ROC) curve comparison between the Soft Voting and Stacking Meta-Learner models, illustrating their effectiveness in binary classification. Both curves hug the top-left corner of the plot, which indicates exceptionally high true positive rates and very low false positive rates.

The area under the curve (AUC) values reinforce this observation Soft Voting achieves an AUC of 0.9992, while Stacking slightly surpasses it with an AUC of 0.9993. The near-perfect AUC values for both models confirm their strong discriminative power, with stacking demonstrating a marginal advantage in overall classification performance. The ROC curve thus effectively visualizes how confidently each model distinguishes between benign and attack traffic in the DDoS Flood Attacks.

Table 4 illustrates a comparative analysis of system-level performance metrics detection time, latency, and computational cost for three federated ensemble learning strategies: EFL (Per Round Update), EFL (Soft Voting), and EFL (Stacking), with the corresponding graphical representation shown in Fig. 11. EFL (Per Round Update) demonstrates the highest resource consumption, with significantly elevated detection time and latency, indicating greater processing overhead. In contrast, Proposed Hybrid - EFL (Soft Voting) reduces these times substantially while maintaining efficiency, showing the benefit of probability-based aggregation. Proposed Hybrid EFL- (Stacking) further improves performance, achieving the lowest detection time and latency among the three, along with the most optimized computational cost. This progression highlights the trade-offs between model complexity and system efficiency, positioning EFL (Stacking) as the most resource-conscious and responsive solution among the approaches evaluated.

9. Practical engineering relevance and deployment considerations

Although full-scale deployment of the Hybrid EFL framework is beyond the scope of the present research, the simulation environment was carefully designed to reflect realistic engineering constraints. The use of the NCSRD DS 5G DDoS dataset, which was collected from actual edge devices such as Raspberry Pi 4 units, industrial grade 5G routers, and WiFi 6 hotspots, ensures that the modeled attack scenarios and traffic patterns closely resemble those found in operational 5G infrastructures.

In order to bridge the gap between theoretical development and practical implementation, this research presents a realistic pathway for deploying the Hybrid EFL framework within engineering environments. The compact model size of 62.64 Mb and the low latency of 190.10 ms make the framework suitable for deployment on resource-constrained edge devices commonly found in smart city infrastructure, industrial Internet of Things systems, and healthcare monitoring platforms. The federated learning strategy employed in this framework involves the transmission of class probabilities rather than raw data, which minimizes bandwidth consumption and enhances data privacy. This methodology conforms to the essential criteria for implementing secure and decentralized systems. In addition, the framework is structured to integrate efficiently with existing intrusion detection infrastructures by employing streamlined application interfaces and modular service containers.This enables modular and scalable deployment across heterogeneous edge nodes.These considerations affirm the engineering applicability of the Hybrid-EFL framework and lay the groundwork for

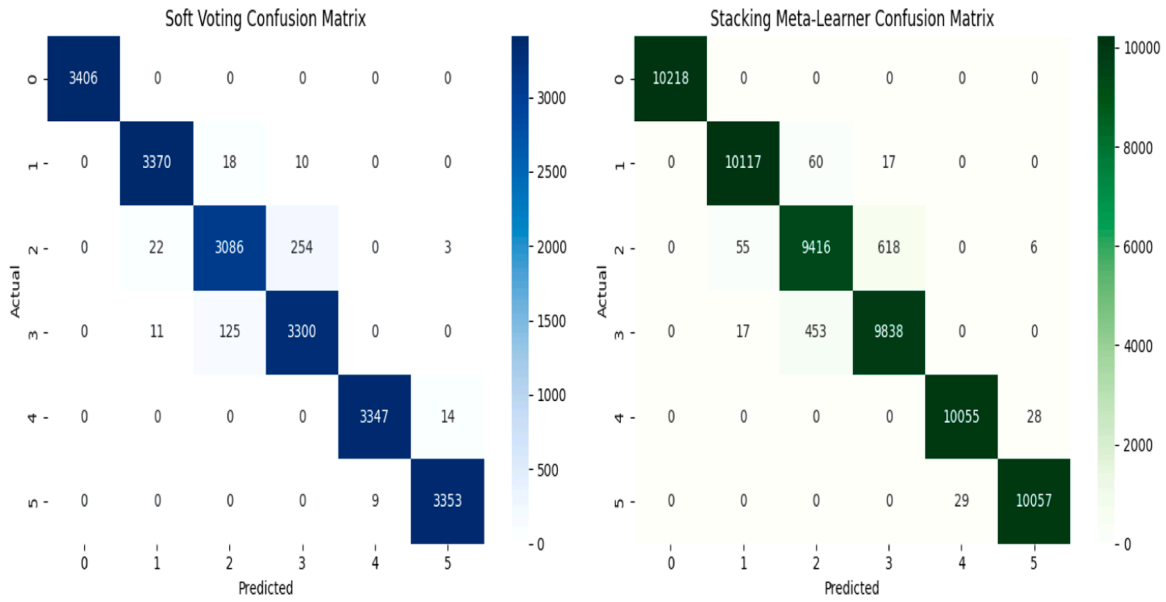


Fig. 8. displays a side-by-side comparison of confusion matrices from two models soft voting and stacking Meta-learner used for classifying DDoS flooding attack types.

Table 3
Comprehensive evaluation of traditional classifiers and ensemble federated learning (EFL) approaches based on key performance metrics.

Model	Accuracy	Precision	Recall	F1-Score	AUC
Random Forest	0.9759	0.9759	0.9759	0.9759	0.9991
Gradient Boosting	0.954	0.9539	0.954	0.9539	0.9978
XGBoost	0.9759	0.9759	0.9759	0.9759	0.9991
EFL (per round)	0.9688	0.9688	0.9688	0.9688	0.9991
Proposed Hybrid EFL Soft Voting	0.9778	0.9778	0.9778	0.9778	0.9992
Proposed Hybrid EFL Stacking	0.979	0.979	0.979	0.979	0.9993

future real-time implementations in 5 G and emerging 6G networks.

10. Conclusion

In conclusion, the Proposed Hybrid - EFL framework demonstrates a robust combination of distributed intelligence, ensemble learning, and resource-efficient design tailored for secure intrusion detection. Among traditional classifiers, Random Forest and XGBoost achieved strong baseline scores with accuracy, precision, recall, and F1-score all at 0.9759 and an AUC of 0.9991, while Gradient Boosting trailed slightly with an accuracy of 0.9540 and AUC of 0.9978. Moving into the federated domain, EFL (Per Round Update) maintained reliable performance with all core metrics at 0.9688 and AUC 0.9991. Performance improved notably with Hybrid - EFL (Soft Voting), which reached 0.9778 on all major metrics and AUC 0.9992**.** Leading the pack, Hybrid - EFL (Stacking**)** achieved the highest values: 0.9790 across accuracy, precision, recall, and F1-score, and a peak AUC of 0.9993. From a system

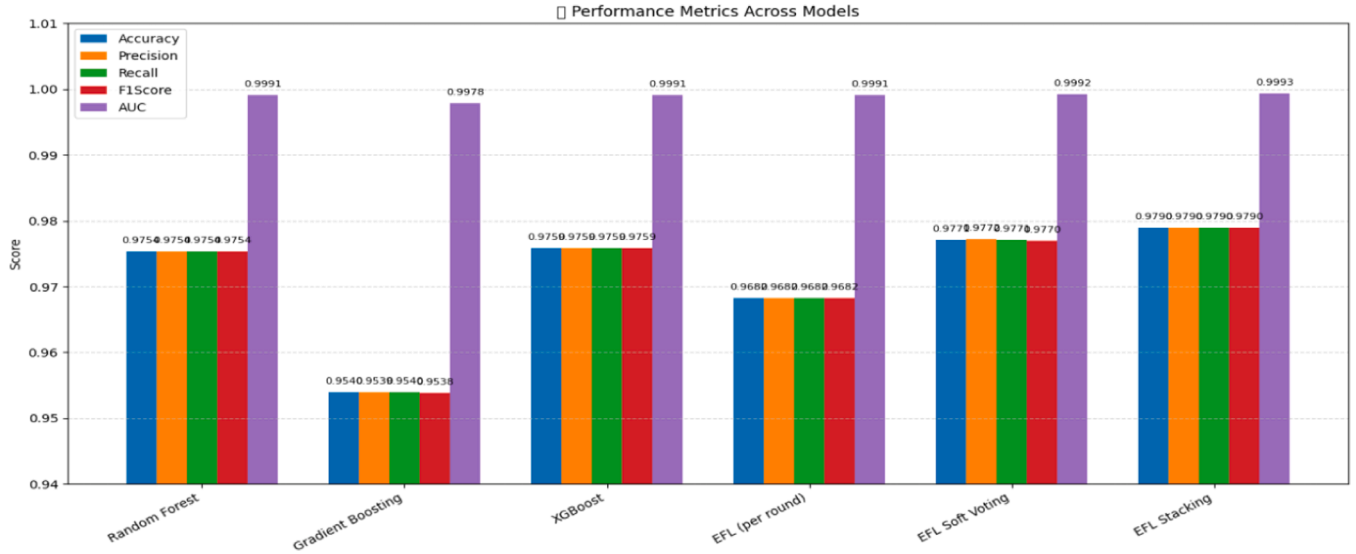


Fig. 9. Graphical comparison of classification performance metrics—accuracy, precision, recall, F1-score, and AUC—for traditional machine learning models and ensemble federated learning (EFL) techniques, illustrating the incremental gains achieved by soft voting and stacking over individual models.

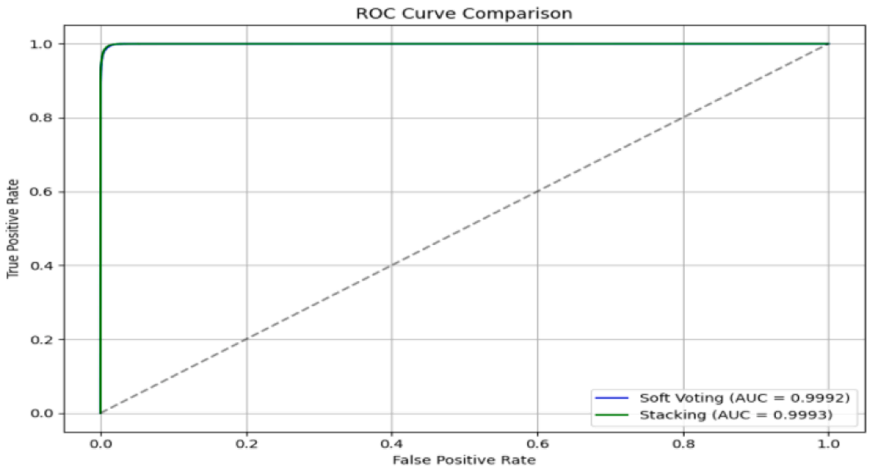


Fig. 10. Receiver operating characteristic (ROC) curve comparison between the proposed hybrid EFL- soft voting and stacking meta-learner models, illustrating their effectiveness in binary classification.

Table 4
Comparative analysis of system-level performance metrics: detection time, latency, and computational cost for three federated ensemble learning strategies: EFL (per round update), proposed hybrid - EFL (soft voting), and EFL (stacking).

Model	Detection Time (ms)	Latency (ms)	Computational cost (MB)
EFL(per round Update)	1376.76	1342.81	112.51
EFL(Soft Voting)	320.48	324.00	62.64
EFL(Stacking)	180.22	190.10	62.64

efficiency perspective, Hybrid - EFL (Stacking) also outperformed the others with 180.22 ms detection time, 190.10 ms latency, and a compact computational cost of 62.64 MB, making it the most optimized choice for real-time network security. These results affirm that the Proposed Hybrid - EFL framework, especially when paired with stacking, offers an effective and scalable solution for privacy-preserving, high-performance intrusion detection.

Future scope

The evolution to 6G networks will bring even greater connectivity, device density, and complexity, further amplifying the need for intelligent, distributed security solutions. Future research can explore the integration of adaptive and self-healing federated learning frameworks capable of responding to novel attack vectors in real-time. The use of advanced optimization techniques, such as dynamic neural architecture search and meta-learning, may further enhance model efficiency for ultra-constrained devices. Additionally, the incorporation of blockchain for federated model integrity, and the extension to multi-modal and cross-domain anomaly detection, will be vital as edge computing environments become more heterogeneous and interconnected. Continuous benchmarking against emerging threats and real-world deployments will ensure that DDoS detection frameworks remain resilient and relevant in the face of rapidly changing cyber landscapes.

CRediT authorship contribution statement

Lakshmi V: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sujatha Rajkumar:** Validation, Supervision.

Declaration of competing interest

All authors associated with this manuscript confirm the following: No direct financial interests exist, such as employment, stock ownership, grants, or patents related to the content of the manuscript. No indirect financial interests exist, such as honoraria, consultancies, mutual fund ownership, or paid expert testimony. No personal relationships, academic competition, or intellectual beliefs are present that could be construed as potential or actual competing interests. This declaration is

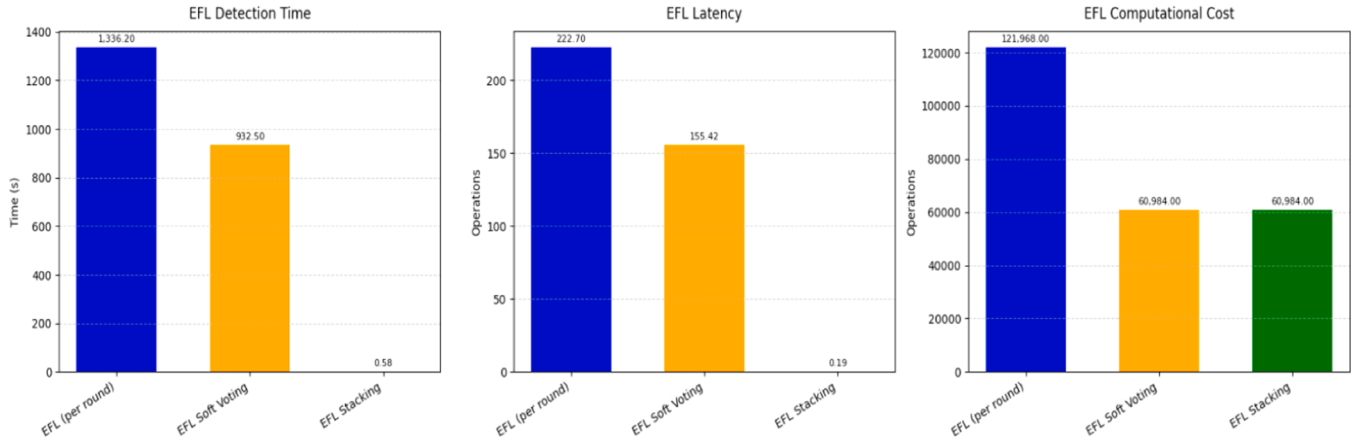


Fig. 11. Visualization of system-level performance metrics detection time, latency, and computational cost—across three ensemble federated learning strategies: EFL (per round update), proposed hybrid- EFL (soft voting), and EFL (stacking), highlighting the efficiency gains achieved through model aggregation.

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Data availability

NCSRD. (2025). NCSRD-DS-5GDDos: A 5G DDoS Attack Dataset. Zenodo. https://doi.org/10.5281/zenodo.13900057.

References

[1] G.A. Akpakwu, B.J. Silva, G.P. Hancke, A.M. Abu-Mahfouz, A survey on 5G networks for the Internet of Things: communication technologies and challenges, IEEE Access. 6 (2017) 3619–3647.

[2] N.B. Henda, Overview on the security in 5G phase 2, J. ICT Stand. 8 (1) (2020) 1–14, https://doi.org/10.13052/jicts2245-800X.811.

[3] Y. Zhang, B. Long, A review of 5G-advanced service and system aspects standardization in 3GPP, in: *2022 IEEE/CIC International Conference on Communications in China (ICCC Workshops)*. IEEE, 2022, https://doi.org/10.1109/ICCCWorkshops55477.2022.9896670.

[4] P.D. Rosero-Montalvo, P. Tözün, W. Hernandez, Optimized CNN architectures benchmarking in hardware-constrained edge devices in IoT environments, IEEe Internet. Things. J. (2024), https://doi.org/10.1109/JIOT.2024.3369607.

[5] M.J. Khan, R.C.S. Chauhan, I. Singh, Z. Fatima, G. Singh, Mobility management in heterogeneous network of vehicular communication with 5g: current status and future perspectives, IEEe Access. 12 (2024) 86271–86292.

[6] R. Shameli, S. Rajkumar, Modified convolutional neural network (MCNN) for intelligent 5G radio software-defined networking (SDN). Design and Optimization of Wearable, Implantable, and Edible Antennas, IGI Global, 2024, pp. 418–442.

[7] M.F. Saiyed, I. Al-Anbagi, Deep ensemble learning with pruning for DDoS attack detection in IoT networks, IEEE Trans. Mach. Learn. Commun. Netw. (2024), https://doi.org/10.1109/TMLCN.2024.3395419.

[8] I. Ortega-Fernandez, M. Sestelo, J.C. Burguillo, et al., Network intrusion detection system for DDoS attacks in ICS using deep autoencoders, Wirel. Netw. 30 (2024) 5059–5075, https://doi.org/10.1007/s11276-022-03214-3APA.

[9] Y. Alhasawi, S. Alghamdi, Federated learning for decentralized DDoS attack detection in IoT networks, IEEE Access. 12 (2024) 42357–42368.

[10] Y.C. Lee, W.C. Chien, Y.C. Chang, FedDB: a federated learning approach using DBSCAN for DDoS attack detection, Appl Sci 14 (3) (2024) 1–15.

[11] A.A. Alashhab, M.S. Zahid, B. Isyaku, A.A. Elnour, W. Nagmeldin, A. Abdelmaboud, U.D. Maiwada, Enhancing DDoS attack detection and mitigation in SDN using an ensemble online machine learning model, IEEE Access. 12 (2024) 51630–51649.

[12] E.F. Swana, W. Doorsamy, P. Bokoro, Tomek link and SMOTE approaches for machine fault classification with an imbalanced dataset, Sensors 22 (9) (2022) 3246.

[13] W. Chandra, B. Suprihatin, Y. Resti, Median-KNN Regressor-SMOTE-Tomek links for handling missing and imbalanced data in air quality prediction, Symmetry 15 (4) (2023) 887.

[14] B. Zhang, Y. Zhang, X. Jiang, Feature selection for global tropospheric ozone prediction based on the BO-XGBoost-RFE algorithm, Sci. Rep. 12 (1) (2022) 9244.

[15] MISBAH, A., Anass S., and I. Hafidi. "“latex SmartEdge: a scalable and adaptive framework for cyber resilience in IoMT." (2024).

[16] I. Cloudflare, DDoS threat reports, Cloudflare Blog (2025). https://blog.cloudflare.com/tag/ddos-reports/.

[17] M. Šimon, L. Huraj and D. Hrinkino, "Home Assistant platform under DDoS attacks for IPv4 and IPv6 networks," *2024 IEEE 17th International Scientific Conference on Informatics (Informatics)*, Poprad, Slovakia, 2024, pp. 402–407, doi: 10.1109/Informatics62280.2024.10900854.

[18] L. De Pedro, A.M. Rosu, J.E..D. Vergara, Server load estimation by Burr distribution mixture analysis of TCP SYN response time, J. Netw. Comput. Appl. 218 (2023) 103694.

[19] Y. Wang, J. Ding, T. Zhang, Y. Xiao, X. Hei, From replay to regeneration: recovery of UDP flood network attack scenario based on SDN, Mathematics 11 (8) (2023) 1897.

[20] M.E. Sahin, M. Demirci, HELP4DNS: leveraging the programmable data plane for effective and robust defense against DDoS attacks on DNS, J. Netw. Comput. Appl. (2025) 104198.

[21] C. Kemp, C. Calvert, T.M. Khoshgoftaar, J.L. Leevy, An approach to application-layer DoS detection, J. Big. Data 10 (1) (2023) 22, https://doi.org/10.1186/s40537-023-00699-3.

[22] J. Ma, W. Su, Collaborative DDoS defense for SDN-based AIoT with autoencoder-enhanced federated learning, Inf. Fusion. 117 (2025) 102820.

[23] W. Lei, A.C.K. Soong, L. Jianghua, W. Yong, B. Classon, W. Xiao, D. Mazzarese, Z. Yang, T. Saboorian, 5G System Design, Springer Science and Business Media LLC, 2021.

[24] B. Bousalem, V.F. Silva, R. Langar, S. Cherrier, DDoS attacks detection and mitigation in 5G and beyond networks: a deep learning-based approach, in: GLOBECOM 2022-2022 IEEE Global Communications Conference, 2022, pp. 1259–1264. IEEE.

[25] N.N.M. Yusof, N.S. Sulaiman, Cyber attack detection dataset: a review, J. Phys. Conf. Ser. 2319 (1) (2022), https://doi.org/10.1088/1742-6596/2319/1/012029. IOP Publishing.

[26] J. Li, L. Lyu, X. Liu, X. Zhang, X. Lyu, FLEAM: a federated learning empowered architecture to mitigate DDoS in industrial IoT, IEEE Trans. Ind. Inform. 18 (6) (2021) 4059–4068.

[27] H. Ahmetoglu, R. Das, A comprehensive review on detection of cyber-attacks: data sets, methods, challenges, and future research directions, Internet Things 20 (2022) 100615 egateway.vit.ac.in/10.1016/j.iot.2022.100615.

[28] D. Lv, X. Cheng, J. Zhang, W. Zhang, W. Zhao, H. Xu, DDoS attack detection based on CNN and federated learning, in: 2021 Ninth International Conference on Advanced Cloud and Big Data (CBD), 2022, pp. 236–241. IEEE.

[29] D. Tymoshchuk, O. Yasniy, M. Mytnyk, N. Zagorodna, V. Tymoshchuk, Detection and classification of DDoS flooding attacks by machine learning method, arXiv preprint (2024) arXiv:2412.18990.

[30] R. Doriguzzi-Corin, D. Siracusa, FLAD: adaptive federated learning for DDoS attack detection, Comput. Secur. 137 (2024) 103597.

[31] Y.S.N. Fotse, V.K. Tchendji, M. Velempini, Federated learning based DDoS attacks detection in large scale software-defined network, IEEE Trans. Comput. (2024).

[32] J. Wang, Y. Si, Y. Zhu, K. Zhang, S. Yin, B. Liu, Cyberattack detection for electricity theft in smart grids via stacking ensemble GRU optimization algorithm using federated learning framework, Int. J. Electr. Power Energy Syst. 157 (2024) 109848.

[33] J. Shen, W. Yang, Z. Chu, J. Fan, D. Niyato, K.Y. Lam, Effective intrusion detection in heterogeneous Internet-of-Things networks via ensemble knowledge distillation-based federated learning, in: In ICC 2024-IEEE International Conference on Communications, 2024, pp. 2034–2039. IEEE.

[34] S. Arroyo, Federated Learning Based Autoencoder Ensemble System for Malware Detection On Internet of Things Devices. MS Thesis, Rowan University, 2024.

[35] N. Jeffrey, Q. Tan, J.R. Villar, Using ensemble learning for anomaly detection in cyber–physical systems, Electronics 13 (7) (2024) 1391.

[36] I. Ullah, X. Deng, X. Pei, H. Mushtaq, Z. Khan, Securing internet of vehicles: a blockchain-based federated learning approach for enhanced intrusion detection, Clust.. Comput 28 (4) (2025) 256.

[37] S.M.S. Bukhari, M.H. Zafar, M. Abou Houran, Z. Qadir, S.K.R. Moosavi, F. Sanfilippo, Enhancing cybersecurity in Edge IIoT networks: an asynchronous federated learning approach with a deep hybrid detection model, Internet Things 27 (2024) 101252.

[38] A.V. Nagarjun, S. Rajkumar, Exploring the potential of deep learning and blockchain for intrusion detection systems: a comprehensive review, J. Circuits Syst. Comput. 33 (16) (2024) 2430007.

[39] M.R. AT, B. Balaji, S.A.P. RR, R.C. Naidu, R.K. M, P. Ramachandran, S. Rajkumar, V.N. Kumar, G. Aggarwal, A.M. Siddiqui, Intelligent energy management across smart grids deploying 6G IoT, AI, and blockchain in sustainable smart cities, IoT 5 (3) (2024) 560–591.

[40] NCSRD, NCSRD-DS-5GDDos: a 5G DDoS attack dataset, Zenodo (2025), https://doi.org/10.5281/zenodo.13900057.

[41] V. Bytyqi, B. Rexha, Machine learning boosted trees algorithms in cybersecurity: a comprehensive review, in: Future of Information and Communication Conference, 2024, pp. 158–173. Cham: Springer Nature Switzerland.